EE 354/CE 361/MATH 310 - Introduction to Probability and Statistics

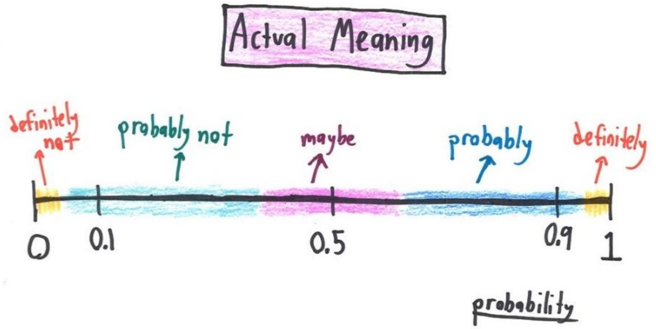
Fall 2025

EE 354/CE 361/MATH 310 L2 section (Fall 2025)

Introduction to Probability and Statistics -

aamir hasan

Week 15 - Lecture No 28 – November 26, 2025



Announcements!

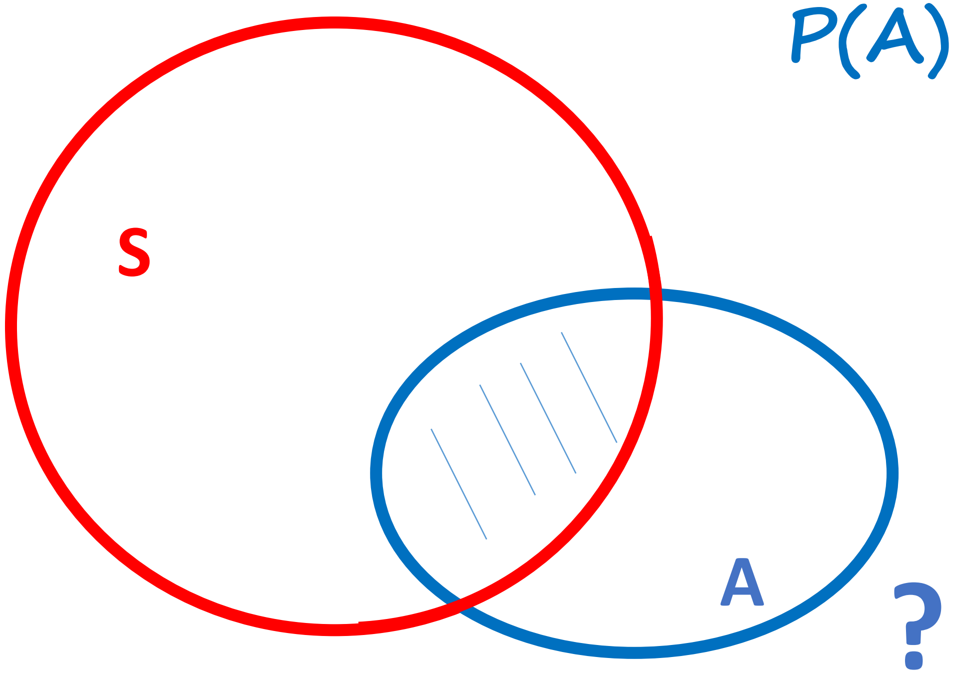
1. All lectures posted.
2. *Assignment 3 posted*
3. **Quiz No 10 – December 03, 2025**

Agenda for today

1. Unit 5 – Continuous Random Variables
   - Joint PDFs
     - **Uniform joint PDF on a set S - Recap**
     - Joint CDF - recap

Uniform joint PDF on a set S

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{area\ of\ S}, & if\ (x,y) \in S, \\ 0, & Otherwise \end{cases}$$



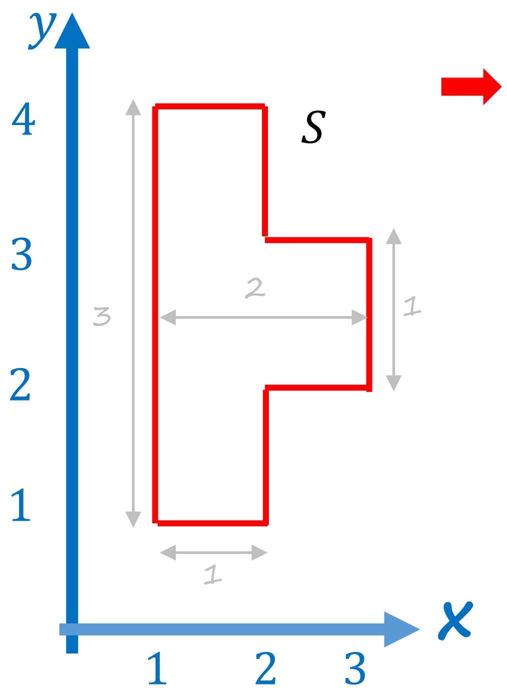
$$P(A) = \frac{area\ (A \cap S)}{area\ (S)}$$

Exercise - Uniform joint PDF on a set S

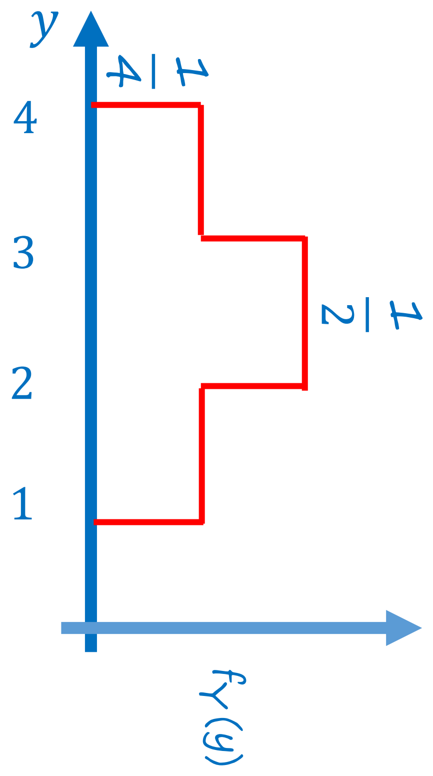
$$f_X(x) = \int f_{X,Y}(x,y)\, dy \qquad f_Y(y) = \int f_{X,Y}(x,y)\, dx$$

$$f_{X,Y}(x,y) = \begin{cases} \dfrac{1}{area\ of\ S}, & if\ (x,y) \in S, \\ 0, & \text{Otherwise.} \end{cases}$$

$\rightarrow f_{X,y} = ?$



(e) $f_Y(y) = ?$

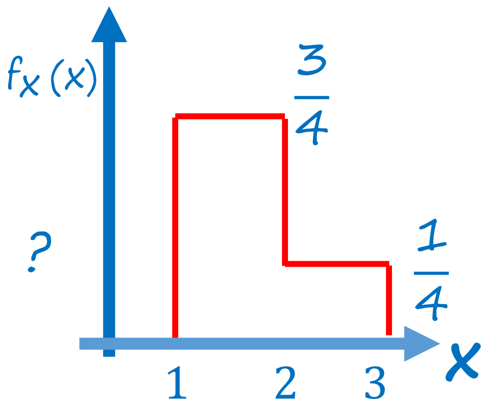


(a) $f_{x,y}(4, 4) = ?$

(b) $f_{x,y}(2, 2) = ?$

(c) $P(1 < X < 2, 1 < Y < 2) = ?$

(d) $f_X(x) = ?$



More than two random variables

- $p_{X,Y,Z}(x, y, z)$
- $\sum_x \sum_y \sum_z p_{X,Y,Z}(x, y, z) = 1$
- $p_X(x) = \sum_z \sum_y p_{X,Y,Z}(x, y, z)$
- $p_{X,Y}(x, y) = \sum_z p_{X,Y,Z}(x, y, z)$

- $f_{X,Y,Z}(x, y, z)$

Sum becomes integrals & pmfs becomes pdfs

Functions of multiple random variables

➡ $Z = g(X, Y)$

Expected value rule:

➡ $E[g(X,Y)] = \sum_x \sum_y g(x,y)\, p_{X,Y}(x,y)$ $\quad E[g(X,Y)] = \int \int g(x,y) f_{X,Y}(x,y) dxdy$

Linearity of expectations

➡ $E[aX + b] = aE[X] + b$

➡ $E[X + Y] = E[X] + E[Y]$

$$\boldsymbol{E[X_1 + \ \dots + X_n] = E[X_1] + \ \dots + E[X_n]}$$

The joint CDF

➡ $F_X(x) = P(X \leq x) = \int_{-\infty}^{x} f_X(t)\,dt$ ➡ $f_X(x) = \dfrac{dF_X(x)}{dx}$

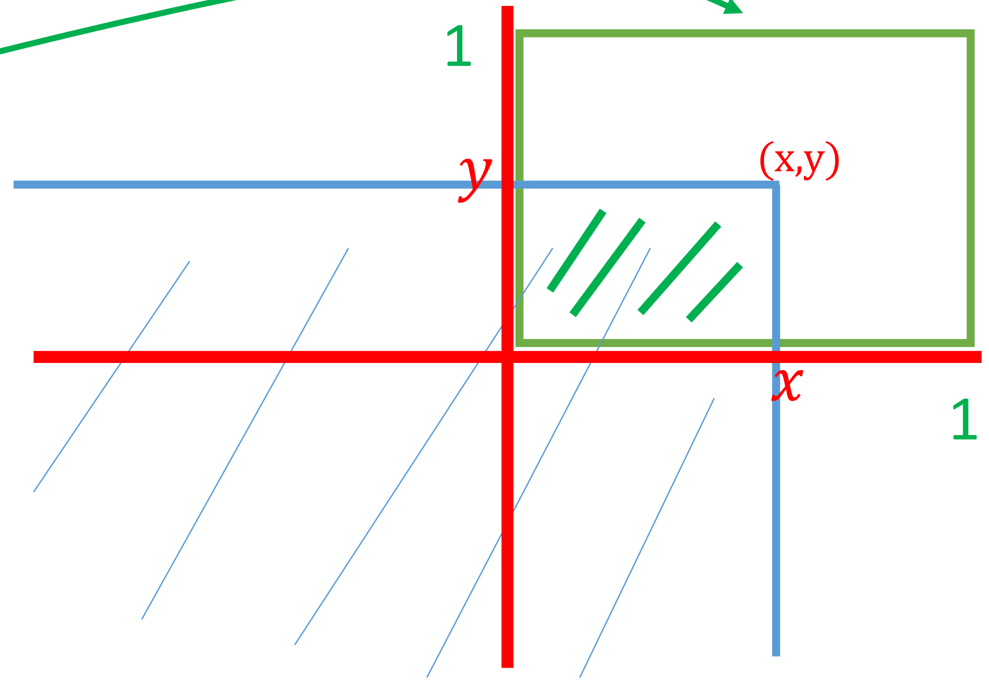
➡ $F_{X,Y}(x,y) = P(X \leq x, Y \leq y\,)$ $= \int_{-\infty}^{y}\left[\int_{-\infty}^{x} f_{X,Y}(s,t)\,ds\right]dt$

➡ $f_{X,Y}(x,y) = \dfrac{\partial^2 F_{X,Y}}{\partial x \partial y}(x,y)$

$f_{X,Y}(x,y) = 1$, for 0 < (x, y) < 1

$F_{X,Y}(x,y) = xy$

$f_{X,Y}(x,y) = 1$



Conditional PDFs, given another r.v.

$$p_{X|Y}(x|y) = \mathbf{P}(X = x \mid Y = y) = \frac{p_{X,Y}(x,y)}{p_Y(y)} \qquad P_y(y)>0$$

$$p_{X,Y}(x,y) \longrightarrow f_{X,Y}(x,y)$$

$$p_{X|A}(x) \longrightarrow f_{X|A}(x)$$

$$p_{X|Y}(x|y) \longrightarrow f_{X|Y}(x|y)$$

$$\boxed{f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \text{if } f_y(y)>0}$$

$$\mathbf{P}(\mathrm{x} \le \mathrm{X} \le x + \delta \mid A) \approx f_{X\mid\mathrm{A}}(x)\,\delta \qquad \text{where } \mathrm{P(A)>0}$$

If $Y = y$; $P(Y=y) = 0$ So $Y \approx y$

$$\mathbf{P}(\mathrm{x} \le \mathrm{X} \le x + \delta \mid \mathrm{y} \le \mathrm{Y} \le y + \Delta) = \frac{f_{X,Y}(x,y)\delta\Delta}{f_Y(y)\ \Delta} = f_{X|Y}(x|y)\,\delta$$

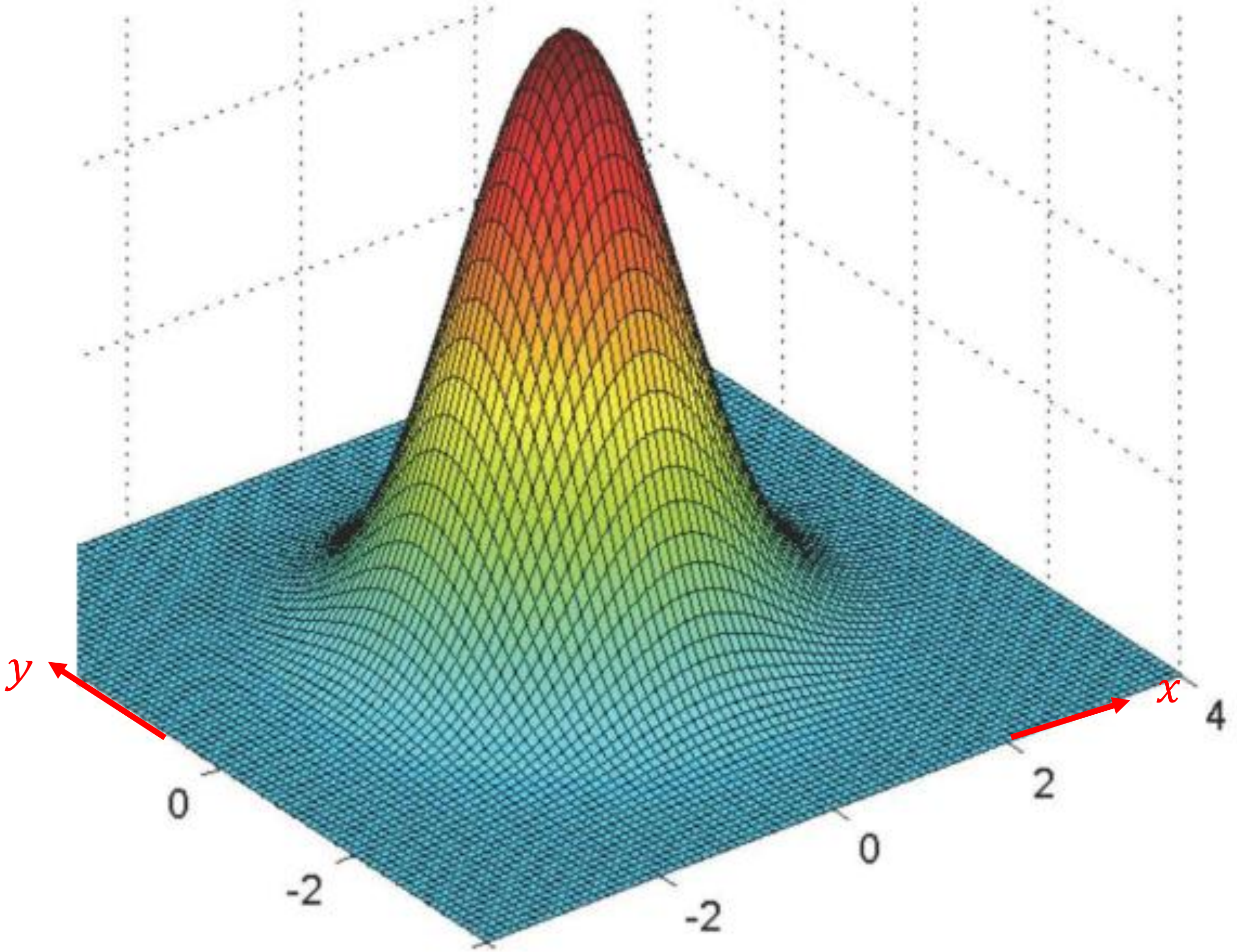
Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)\,dx$$

- Some comments:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

  - $f_{X|Y}(x \mid y) \geq 0$



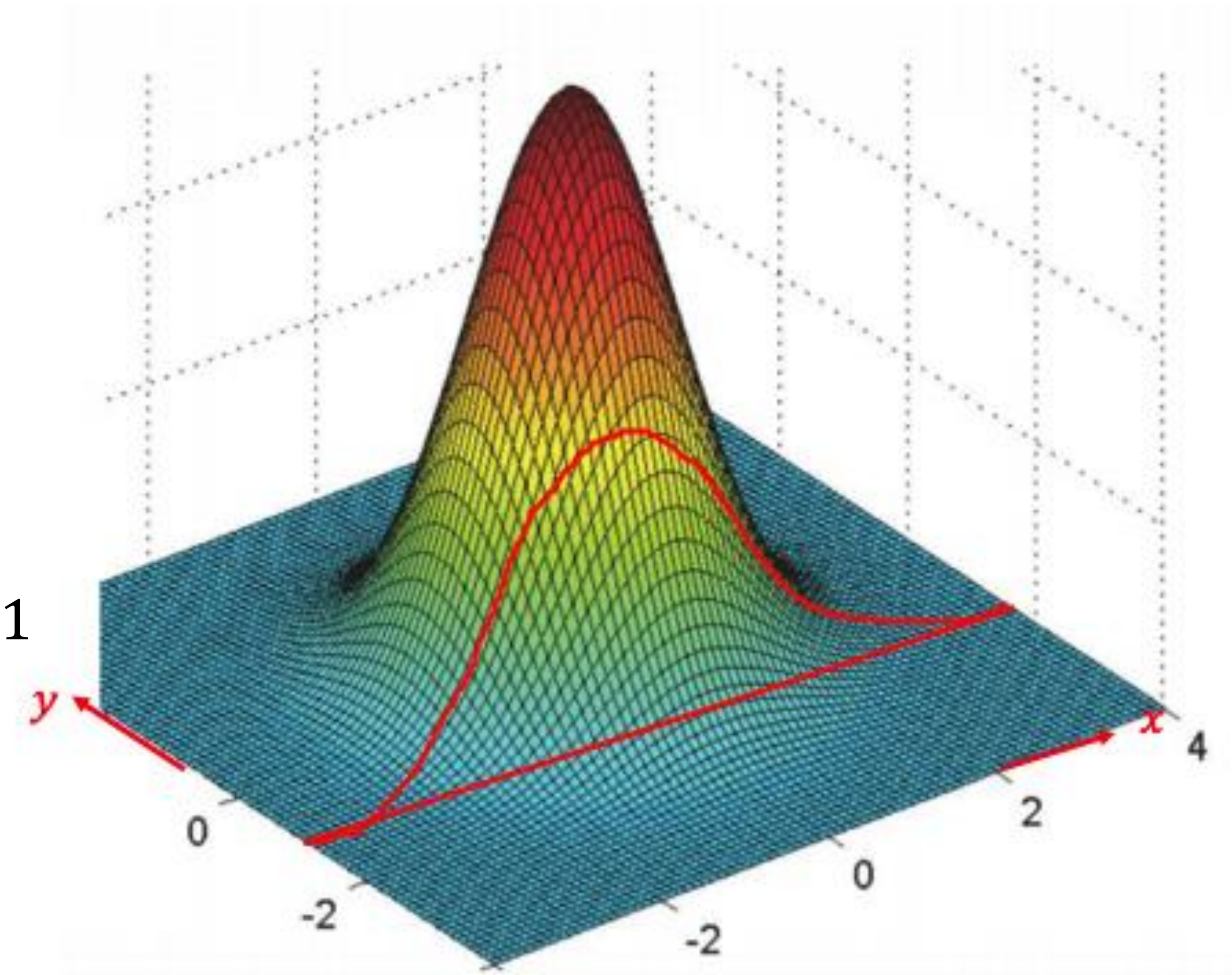
Conditional PDFs, given another r.v.

$$\mathbf{P}(X \in B \mid Y = y) = \int_B f_{X|Y}(x|y)\,dx$$

- **Some comments:**

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

  - $f_{X|Y}(x \mid y) \geq 0$
  - Think of the value of Y fixed at y
    shape of $f_{X|Y}(.|y)$: slice of the joint
  - $\int_{-\infty}^{\infty} f_{X|Y}(x|y)\mathrm{d}x = \frac{\int_{-\infty}^{\infty} f_{X,Y}(x,y)\mathrm{d}x}{f_Y(y)} = 1$
  - Multiplication rule:
    $f_{X,Y}(x,y) = f_{X|Y}(x|y)\,.\,f_Y(y)$
    $= f_{Y|X}(y|x)\,.\,f_X(x)$



Expected Value Rule

$$\mathbf{E}[\, g(X) \mid Y = y\,] = \sum_x g(x)\, p_{X|Y}(x \mid y)$$

$$\mathbf{E}\left[g(X) \middle| Y=y\right] = \int_{-\infty}^{\infty} g(x)\, f_{X|Y}(x|y)\, dx$$

Independence

$$p_{X,Y}(x,y) = p_X(x)\ p_Y(y)$$ for all x, y

$$f_{X,Y}(x,y) = f_X(x)\ f_Y(y)$$ for all x, y

$$f_{X,Y}(x,y) = f_{X|Y}(x|y).\ f_Y(y)$$ always true

$$\Rightarrow \quad f_X(x) = f_{X|Y}(x|y)$$ for all y with f(y) > 0 and all x

$$\Rightarrow \quad f_Y(y) = f_{Y|X}(y|x)$$ for all x with f(x) > 0 and all y

Total probability & expectation theorem

$$p_X(x) = \sum_y p_Y(y)\, p_{X|Y}(x|y) \longrightarrow f_X(x) = \int_{-\infty}^{\infty} f_Y(y)\, f_{X|Y}(x|y)\,dy$$

$$\mathbf{E}[\, X \mid Y = y\,] = \sum_x x\, p_{X|Y}(x \mid y) \longrightarrow E[\, X \mid Y=y \,] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx$$

$$\mathbf{E}[\, X\,] = \sum_y p_Y(y)\, \mathbf{E}[\, X \mid Y = y\,] \longrightarrow E[\, X \,] = \int_{-\infty}^{\infty} f_Y(y)\, E[\, X \mid Y=y \,]\, dy$$

How do you prove the last statement?

Think → Discuss → Prove

Proof

$$\mathbf{E}[\,X\,] = \int_{-\infty}^{\infty} f_Y(y)\,\mathbf{E}[\,X \mid Y = y\,]\,\mathrm{d}y = \int_{-\infty}^{\infty} f_Y(y) \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx\,dy$$

$$= \int_{-\infty}^{\infty}\int_{-\infty}^{\infty} x\, f_Y(y)\, f_{X|Y}(x|y)\,dy\,dx$$

$$= \int_{-\infty}^{\infty} x \underline{\int_{-\infty}^{\infty} f_Y(y)\, f_{X|Y}(x|y)\,dy}\,dx$$

$$= \int_{-\infty}^{\infty} x f_X(x)\,dx = \mathbf{E}[\,X\,]$$

Example

Uniform probabilities on a triangle

find:

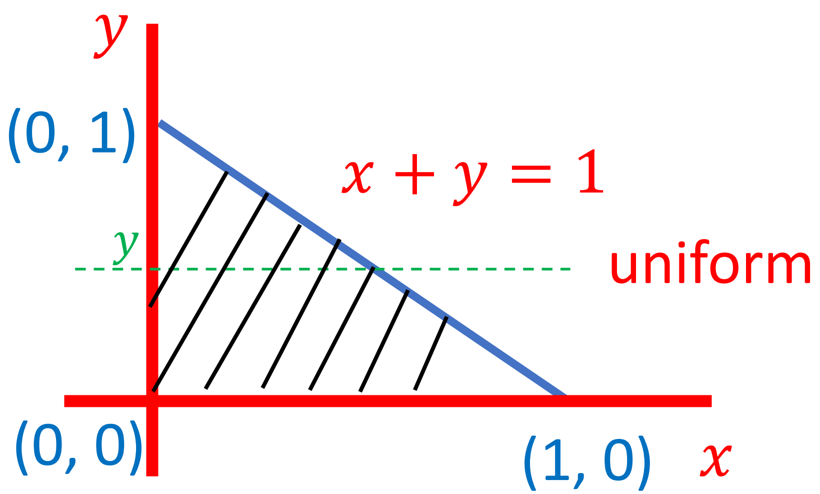
a) $f_{X,Y}(x, y)$

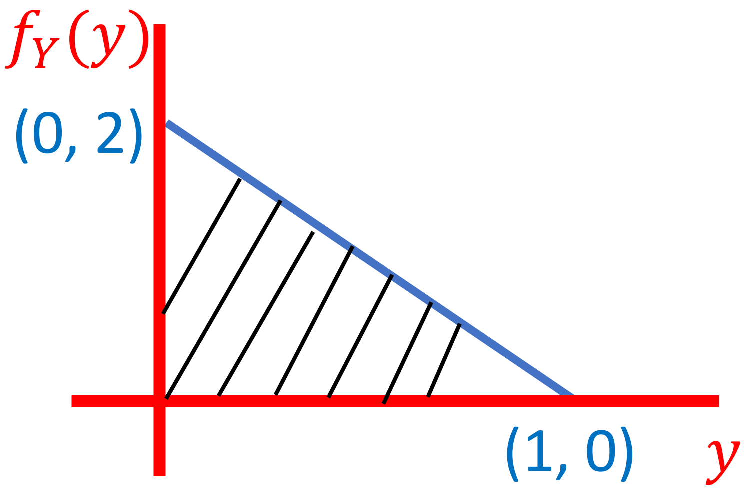
$$f_{X,Y}(x, y) = \begin{cases} 2, & x + y < 1, \\ & x, y > 0 \\ 0, & \text{otherwise} \end{cases}$$

b) $f_Y(y)$

$$f_Y(y) = \int f_{X,Y}(x, y)\, dx$$

$$= \begin{cases} 2(1-y), & 0 \le y \le 1 \\ 0, & \text{otherwise} \end{cases}$$





Example

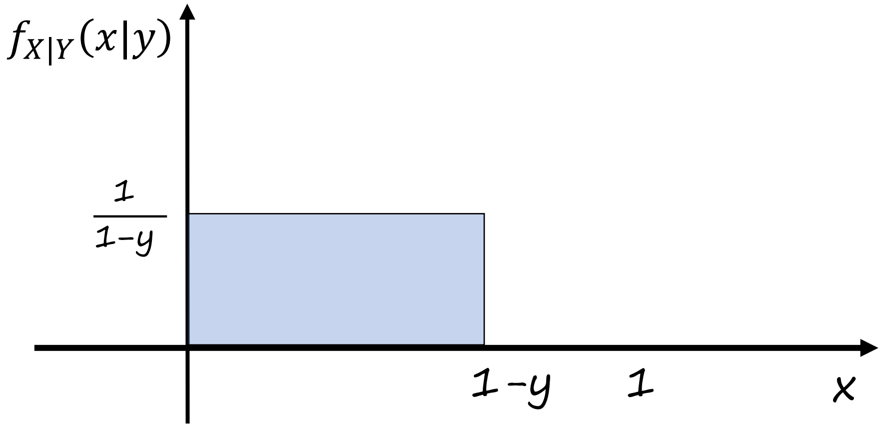
c) $f_{X|Y}(x|y)$

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)}$$

$$= \frac{2}{2(1-y)}$$

$$= \begin{cases} \frac{1}{(1-y)}, & 0 \leq y \leq 1;\ 0 \leq x \leq 1-y \\ 0, & \text{otherwise} \end{cases}$$

$0 \leq y < 1;$
$0 \leq x + y \leq 1$



Example

d) $E[X|Y = y]$

$$E[X \mid Y=y] = \int_{-\infty}^{\infty} x f_{X|Y}(x|y)\,dx$$

$$E[X \mid Y=y] = \int_{0}^{1-y} x \frac{1}{(1-y)}\,dx$$

$$= \frac{1}{(1-y)} \frac{(1-y)^2}{2}$$

$$= \frac{(1-y)}{2} \qquad 0 \le y \le 1$$

Example

e) $E[X]$

$$E[X] = \int_{-\infty}^{\infty} f_Y(y)\, E[X \mid Y=y]\, dy$$

$$= \int_{-\infty}^{\infty} f_Y(y) \frac{(1-y)}{2} dy = \frac{1}{2}\int_{-\infty}^{\infty} f_Y(y) dy - \frac{1}{2}\int_{-\infty}^{\infty} y\, f_Y(y) dy$$

$$= \frac{1}{2} - \frac{1}{2} E[Y] = \frac{1}{2} - \frac{1}{2} E[X]$$

$$\left(1 + \frac{1}{2}\right) E[X] = \frac{1}{2}$$

$$E[X] = \frac{1}{3} \qquad E[Y] = \frac{1}{3}$$

Example on Conditional PDFs

- You have two options to play a game of lottery:

  ❖ Option 1: Select $X \sim U(0, 100)$ then select $Y \sim U(0, X)$. You win Y rupees. Find $f_Y(y)$ and $E[Y]$

  ❖ Option 2: Select $X \sim U(0, 50)$ then select $Y \sim U(0, 2X)$. You win Y rupees. Find $f_Y(y)$ and $E[Y]$

Which option should you select?

Think → Write → Answer

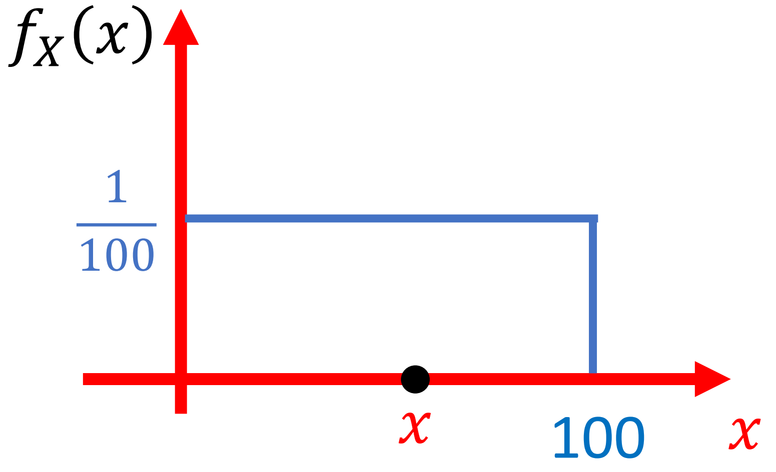
Example on Conditional PDFs

Lets analyze option 1:

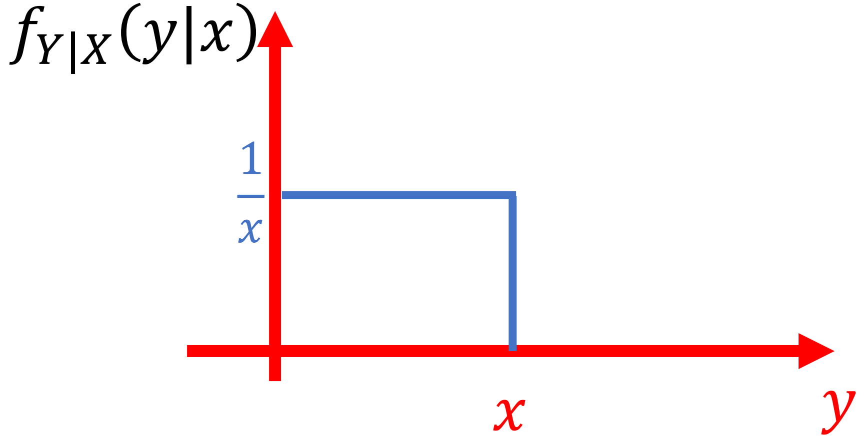
Select x, where $X \sim \mathbf{U}\,(\,\mathbf{0}\,,\mathbf{100}\,)$ then select y such that $Y \sim \mathbf{U}\,(\,\mathbf{0}\,,\mathbf{X}\,)$.
You win y rupees. Find $f_Y(y)$ and $E[Y]$

Option 1: Select x, where $X \sim U(0, 100)$ then select y such that $Y \sim U(0, X)$.
You win y rupees. Find $f_Y(y)$ and $E[Y]$

Are X & Y independent ? If $x = 0.5$, this implies $y \leq 0.5$

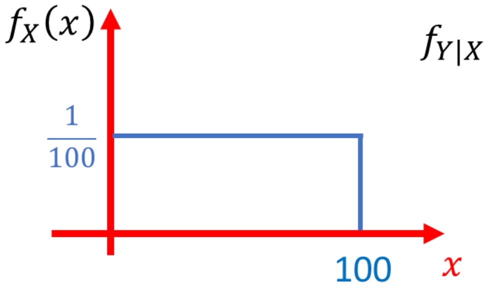


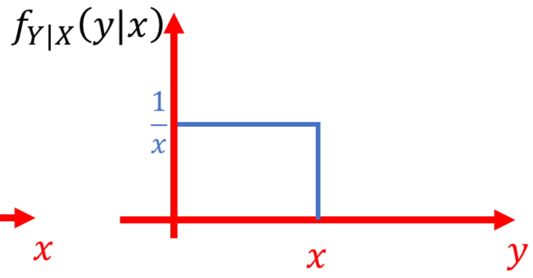
$$f_{X,Y}(x,y) = f_{Y|X}(y|x) . f_X(x)$$



Option 1: Select $X \sim U(0,100)$ then select $Y \sim U(0,X)$. You win Y rupees.

Find $f_Y(y)$ and $E[Y]$



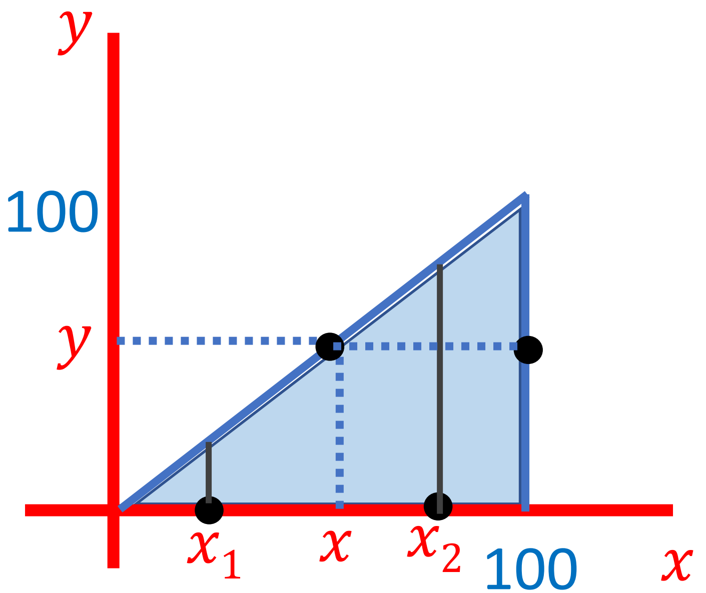


$$f_{X,Y}(x,y) = f_{Y|X}(y|x)f_X(x)$$

$$= \frac{1}{100} * \frac{1}{x} \qquad 0 \le y \le x \le 100$$

$$f_Y(y) = \int_x f_{X,Y}(x,y)dx = \int \frac{1}{100} * \frac{1}{x}\,dx$$

$$= \frac{\ln x}{100}\Big|_y^{100} = \frac{1}{100}\ln(100/y)\ ?$$



$$E[Y] = \int_0^{100} y\frac{1}{100}\ln(100/y)\,dy$$

$$= \int_0^{100} E[Y|X=x]f_X(x)dx = \int_0^{100} \frac{1}{100} * \frac{x}{2}dx = 1/2\ E[X] = 25$$

Option 2: Select $X \sim U(0, 50)$ then select $Y \sim U(0, 2X)$. You win Y rupees.

Find $f_Y(y)$ and $E[Y]$

$$f_{X,Y}(x,y) = f_{Y|X}(y|x) f_X(x)$$

$$= \frac{1}{50} * \frac{1}{2x} = \frac{1}{100x}$$

$0 \le x \le 50;\ 0 \le y \le 2x$

$\Downarrow$

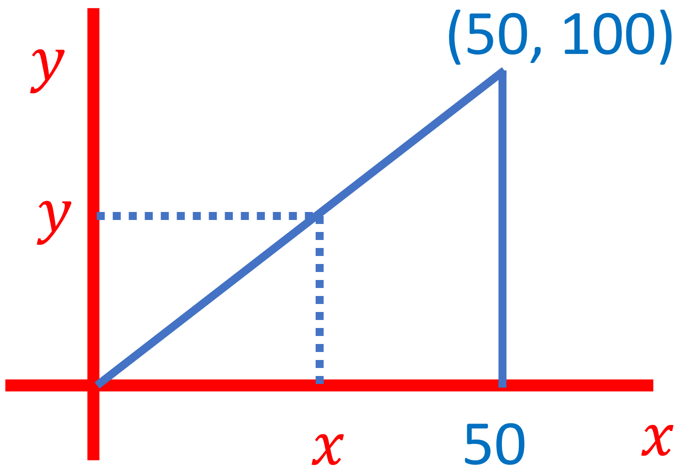
$0 \le y \le 100$

$$f_Y(y) = \int_x f_{X,Y}(x,y)\,dx = \int_{y/2}^{50} f_{X,Y}(x,y)\,dx$$

$$= \left.\frac{\ln x}{100}\right|_{y/2}^{50} = \frac{1}{100}\ln(100/y)$$

$$E[Y] = \int_0^{50} y \frac{1}{100} \ln(100/y)\,dy$$

$$= \int_0^{50} E[Y|X=x] f_X(x)\,dx = \int_0^{50} \frac{1}{50} x\,dx = 1/2\ E[X] = 25$$



Independence

If X and Y are independent

- $\mathbf{E}[\, X\, Y\,] = \mathbf{E}[\, X\,]\, \mathbf{E}[\, Y\,]$
- $\boldsymbol{var}(\mathrm{X} + \mathrm{Y}) = \boldsymbol{var}(\mathrm{X}) + \boldsymbol{var}(\mathrm{Y})$

if g(x) and h(y) are also Indepedent

$\Rightarrow \mathbf{E}[\, g(X)\, h(Y)\,] = \mathbf{E}[\, g(X)\,]\, \mathbf{E}[\, h(Y)\,]$

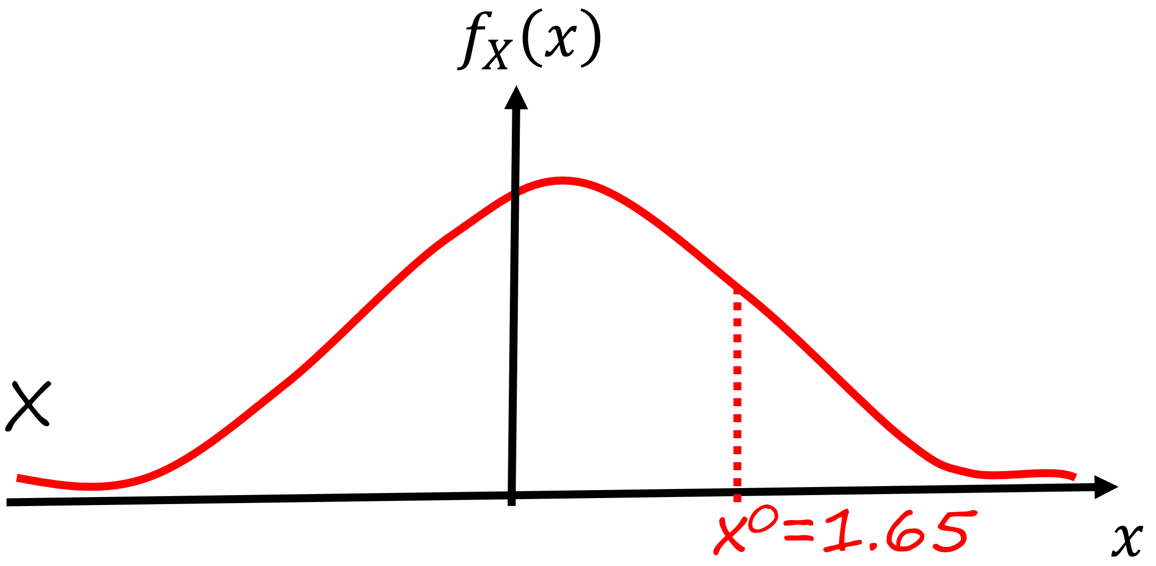
Example:

- Independence in Normal RV's X and Y:

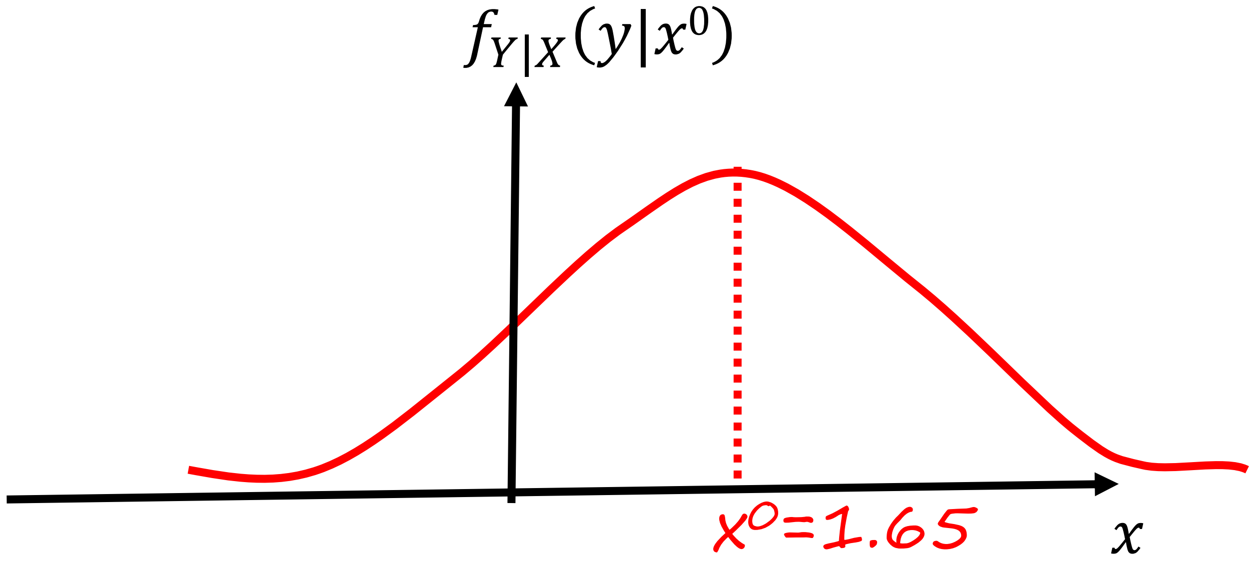
  Lets form a RV in a two step process

1. $X \sim N(0, 1)$;

pick an X



2. $Y \sim N(X, 1)$



are X and Y independent?

Independent Standard Normal

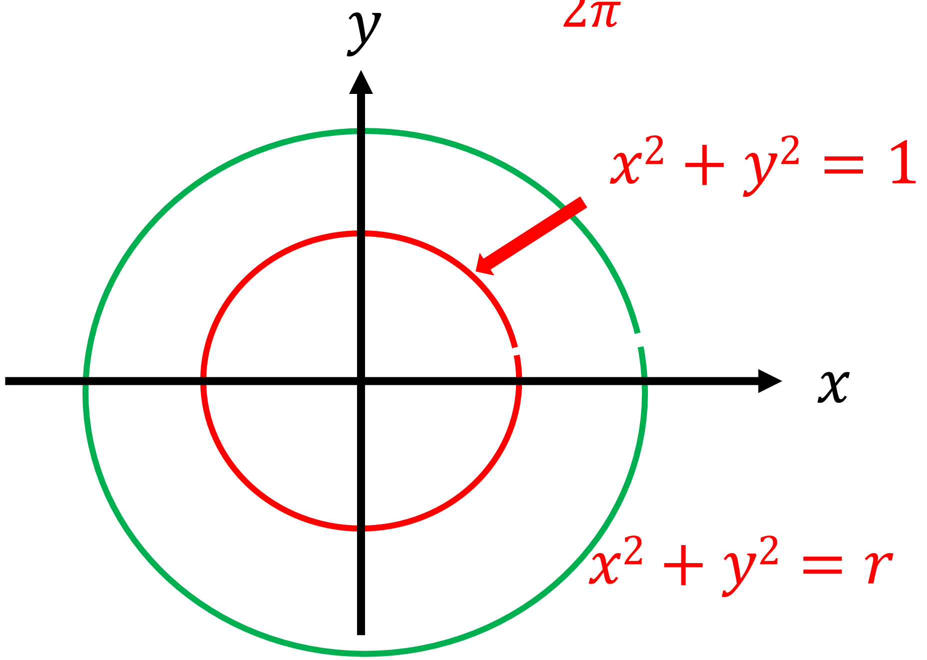
Now let's assume X and Y independent on $\sim N(0, 1)$

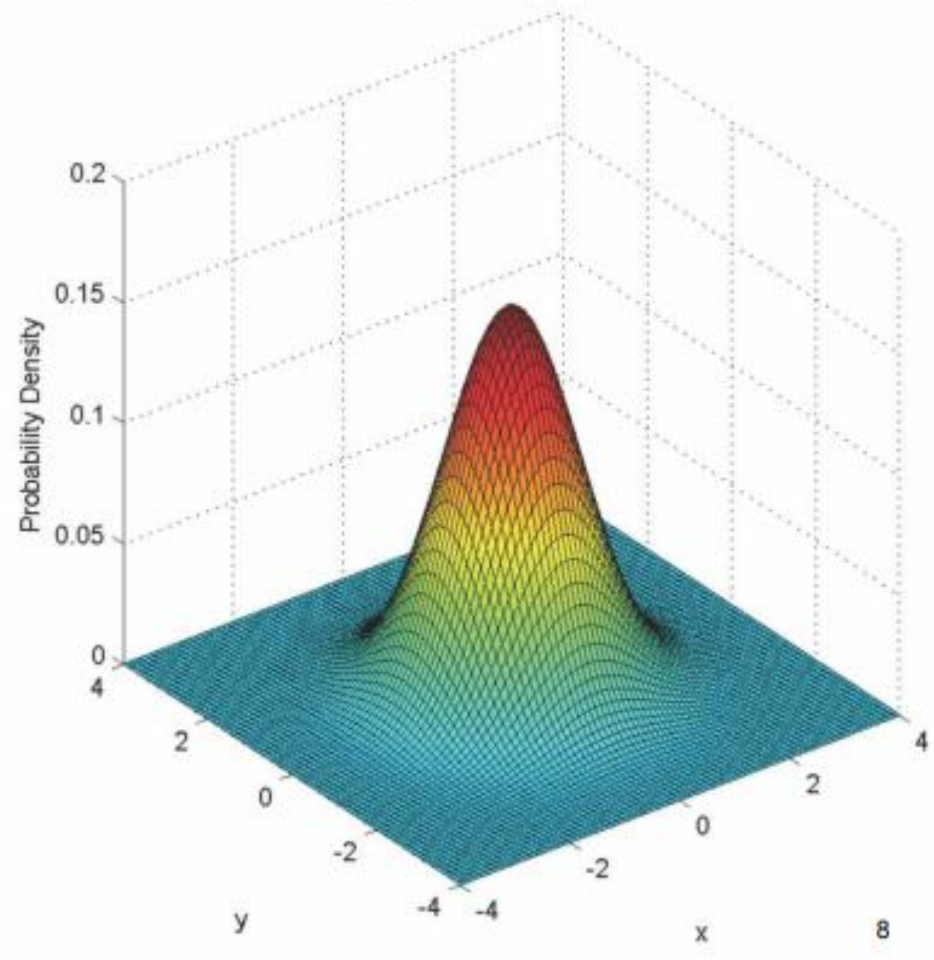
$$\mu_x = \mu_x = 0; \quad \sigma^2{}_x = \sigma^2{}_y = 1;$$

$$f_{X,Y}(x,y) = f_X(x)\ f_Y(y)$$

$$= \frac{1}{\sqrt{2\pi}}\ e^{-\frac{x^2}{2}}\ \frac{1}{\sqrt{2\pi}}\ e^{-\frac{y^2}{2}}$$

$$= \frac{1}{2\pi} \cdot e^{-\frac{1}{2}(x^2+y^2)}$$





Independent Normal

$$f_{X,Y}(x,y) = f_X(x).f_Y(y)$$

$$= \frac{1}{2\pi\sigma_x\sigma_y} e^{\left\{-\frac{(x-\mu_x)^2}{2\sigma_x^2} - \frac{(y-\mu_y)^2}{2\sigma_y^2}\right\}}$$

$\mu_x = \mu_y = 0; \; \sigma^2_x = \sigma^2_y = 4;$